

Sean Chen

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EDUCATION

University of California, Berkeley

B.S. in Mechanical Engineering

Berkeley, CA

August 2026 – May 2028

Mount San Antonio College

A.S. in Mathematics, SASE Co-Founder and Vice President | GPA: 4.0

Walnut, CA

August 2024 – May 2026

- Relevant Coursework: Statics, Mechanics of Material, Material Science, MATLAB

EXPERIENCE

Mechanical Researcher

NASA LSPACE

September 2024 – January 2025

Walnut, CA

- Worked on an interdisciplinary team to develop a NASA proposal based on a decentralized robotic swarm
- Directed research based on system-level requirements for: mechanical hardware (wheels, chassis), thermal control, power systems, and communication systems
- Collaborated with subject matter experts to develop a researched proposal based on NASA taxonomy TX04 (Robotics and Autonomous Systems)

Mechanical Engineer

Pollution Analysis Research

February 2026 – June 2026

Walnut, CA

- Collaborated with a team of 5 students to design a tethered weather balloon system, mapping vertical air quality profiles at altitudes up to 300 meters.
- Designed and fabricated a 3D-printed payload using Fusion 360 to house sensitive sensors and calculated optimal weight-to-lift ratio

California NanoSystems Institute

UCLA

February 2026 – April 2026

Westwood, CA

- Manufactured wafer-based microstructures within a cleanroom environment using photolithography, electron-beam lithography, thin-film deposition, and wet chemical etching.
- Optimized fabrication workflows by adjusting resist thickness and exposure parameters, maximizing pattern fidelity and process repeatability.

PROJECTS

Autonomous Object-Tracking System | *Fusion 360, Raspberry Pi, Python*

October 2025 – February 2026

- Applied principles in Statics to design mechanical structures and joints
- Prototyped and designed a 3-axis motorized gimbal using Fusion 360
- Assisted in developing an object detection algorithm using YOLOv8 for object detection

Posture Pack – Cal Poly Pomona Hackathon | *SolidWorks, Raspberry Pi*

April 2026

- Designed and prototyped a posture-tracking device to help students maintain proper posture while studying
- CAD-model a custom housing to integrate a Raspberry Pi, breadboard, and distance sensor to track the user's position
- Developed a React/TypeScript interface to collect, display, and create a model of posture data for user feedback.

Mazda CX-5 Analysis | *MATLAB*

January 2026

- Used MATLAB to analyze vehicle data, modeling position, velocity, acceleration, RPM, and fuel rate over time
- Created mathematical models and data visualizations from OBD sensor data to evaluate engine and drivetrain behavior

Folding Knife | *SolidWorks, 3D Printing, Fabrication*

April 2026- May 2026

- Designed and prototyped a back lock folding knife in SOLIDWORKS using A2 tool steel
- Enhanced material strength by evaluating material behavior and load paths achieving a 273% increase in material hardness through precise heat treatment

TECHNICAL SKILLS

CAD: SolidWorks, Fusion 360, Siemens NX, Onshape

Software: Python, MATLAB, Java, Arduino, Excel